

SiteWhere: An Open Platform for Connected Devices

SiteWhere delivers a complete management system to help users avoid per device charges by SaaS operators. It connects devices using MQTT, AMQP, STOMP and other protocols, and devices can be added through self-registration, REST services or in batches. It can also control large numbers of devices using batch command operations.



SiteWhere is an open platform for monitoring and controlling IoT devices and storing device data. It is licensed under CPAL 1.0 and can be deployed as an on-premise solution with complete freedom and scalability, which eliminates the per device or per data pricing by SaaS providers. It can also be deployed in a bare metal cloud environment for public access. It comes with advanced multiple tenant architecture to run various IoT applications in a single server instance. It integrates with proven software and services for the enhanced enterprise needs of connected devices. This article familiarises you with the platform, focusing on device communication using JSON payloads over MQTT and on availing platform services using REST APIs.

Architecture and features

The core platform of the server is based on Spring Boot from v1.7.0 and comes with an embedded Tomcat server in recent versions, but you can still build the Web Application Archive (WAR) file from sources, and deploy it in a custom

or vanilla version of the Tomcat server. It comes with multi-tenancy support from v1.2.0, each tenant engine is isolated, having its own data storage and processing channels. It uses MongoDB for the default data store, with *SiteWhere* as the core database and isolated databases for each of the tenants, with tenant-xxxx naming. Devices can communicate using the JSON format or Google Protocol Buffers (Protobuf) over MQTT transport, and REST APIs are offered to avail platform services.

Each tenant device is placed in a collection, known as a site, to cluster the location-aware devices. This enables monitoring of all events at the site level as well as the device level. On bootstrap, a default tenant engine is created with *Construction Site*, initially. Devices can be kept under device groups as logical units; these can associate with device specifications as per common hardware features, and associate with assets for real world mapping.

Components and integration

SiteWhere makes use of the following popular open